

Date: 07 July 2026	Team ID: SB-7429
Project Name: Smart Lender: Applicant Credibility Prediction for Loan Approval	Maximum Marks: 4 Marks

SOLUTION REQUIREMENTS SPECIFICATION

Functional Requirements (FR)

FR ID	Feature Name	Detailed System Requirement Specification
FR-1	Evaluation Form	The system must render a web page capturing 11 features: Gender, Married, Dependents, Education, Self_Employed, ApplicantIncome, CoapplicantIncome, LoanAmount, Loan_Amount_Term, Credit_History, and Property_Area.
FR-2	Imputation Handling	The data pipeline must fill numerical missing values with training means and categorical missing values with modes during training and validation.
FR-3	Scaling & Transforming	The system must map categorical strings to numeric encodings and apply fitted StandardScaler coefficients on numeric fields before running inference.
FR-4	XGBoost Classification	The Flask server must load the serialized XGBoost model (loan_model.pkl) and classify eligibility based on the input vector, outputting probabilities.
FR-5	Decision Panel	The outcomes dashboard must display a clear APPROVED (green) or REJECTED (red) banner, exact probability metrics, and a summary table of inputs.

Non-Functional Requirements (NFR)

NFR ID	Quality Attribute	Specification & Operational Threshold
NFR-1	Performance & Speed	End-to-end inference (form POST, scaling, model prediction, and rendering) must complete in under 120ms.
NFR-2	Usability & Layout	The web dashboard must feature a responsive dark fintech theme supporting navigation tabs, clear typography, and count visualizations.
NFR-3	Model Accuracy	The primary classifier (XGBoost) must meet a testing accuracy of 81.1% on validation splits, balancing precision and recall.
NFR-4	Robustness	Input form must validate variables (e.g. positive values for income) and handle missing inputs gracefully without server crashes.

User Stories & Acceptance Criteria

User Story ID	User Role	As a... I want to... So that...	Acceptance Criteria
US-01	Credit Admin	As a credit admin, I want to instantly screen loan applicants based on demographic and financial inputs, so that I can reduce processing backlog.	Web form accepts 11 parameters, runs prediction using XGBoost, and displays outcome banner in under 120ms.
US-02	Risk Analyst	As a risk analyst, I want to view data distribution charts on the dashboard, so that I can track historical distributions and trends.	Dashboard features an 'Analytics' tab rendering high-resolution count plots and histograms of train data.
US-03	Compliance Auditor	As a compliance officer, I want to review the submitted values on the result page, so that I can verify eligibility rules and compliance.	The result page renders a table showing all applicant attributes (monthly income, credit history, etc.) alongside the final decision.